DATA SCIENCE INTERNSHIP - WEEK 1

Task 1: E-Commerce Dataset Cleaning

Cleaning, Validation and Preparation of Customer Shopping Behavior Data

Organization: WeIntern Pvt Ltd
Tool: Python / Pandas / Jupyter Notebook

1. Objective

The objective of Task 1 is to inspect, clean, validate, and prepare an e-commerce customer shopping behavior dataset for reliable downstream analysis. The workflow focuses on missing values, duplicate records, consistent naming, logical validation, redundant information, and useful derived features while preserving the original raw dataset.

2. Dataset Overview

Item	Value
Dataset	Customer Shopping Behavior
Raw dimensions	3,900 rows x 18 columns
Record identifier	Customer ID
Cleaned dimensions	3,900 rows x 19 columns
Primary tool	Python with Pandas

The dataset contains customer demographics and shopping-behavior fields including age, gender, purchased item, category, purchase amount, location, review rating, subscription status, shipping type, discounts, previous purchases, payment method, and purchase frequency.

3. Initial Data Inspection

The raw dataset was inspected with head(), info(), and describe() to understand dimensions, data types, value ranges, and categorical distributions. The inspection showed that most fields were complete and that Review Rating was the only field with missing observations.

4. Missing Value Analysis and Treatment

Field	Missing Count	Missing %	Decision
Review Rating	37	0.95%	Impute by product-category median

The 37 incomplete Review Rating records were retained because the missing proportion was small and the remaining customer and purchase information was still useful. Missing ratings were filled

with the median Review Rating of the corresponding product category. This preserves all records while using a category-specific typical value rather than a single dataset-wide value.

After imputation, the dataset contained 0 missing values.

5. Column Name Standardization

Column names were standardized to lowercase snake_case for consistent programmatic use. Examples include Customer ID -> customer_id, Item Purchased -> item_purchased, and Review Rating -> review_rating. Purchase Amount (USD) was simplified to purchase_amount.

6. Feature Engineering

Derived Feature	Method	Purpose
age_group	Four quantile-based groups: Young Adult, Adult, Middle-aged, Senior	Supports broader demographic comparison while retaining age.
purchase_frequency_days	Maps purchase-frequency labels to approximate days	Provides a numerical representation for later analysis.

The purchase-frequency mapping uses Weekly = 7, Fortnightly/Bi-Weekly = 14, Monthly = 30, Quarterly/Every 3 Months = 90, and Annually = 365 days. The original categorical frequency field is retained.

7. Data Validation

Validation Rule	Invalid Records
age > 0	0
purchase_amount > 0	0
previous_purchases >= 0	0
Leading/trailing whitespace in categorical fields	0 detected

No records required correction or removal under the documented numerical validation rules. Categorical fields were also checked for leading or trailing whitespace, and no such inconsistencies were detected.

8. Redundant Feature Check

discount_applied and promo_code_used were compared row by row and matched for every record. Because the two fields contained identical information, promo_code_used was removed and discount_applied was retained to avoid redundant features.

9. Duplicate Analysis

Check	Result	Action

Fully duplicated rows	0	No rows removed
Repeated Customer ID values	0; 3,900 unique IDs	No identifier duplicates to resolve

No confirmed duplicate records were found. Consequently, no observations were deleted as duplicates. This preserves the full source record count and follows the principle of removing data only when duplication is confirmed.

10. Final Cleaning Summary

- Preserved all 3,900 source records.
- Resolved 37 missing Review Rating values using category-level median imputation.
- Standardized column names to lowercase snake_case.
- Created age_group and purchase_frequency_days.
- Validated age, purchase amount, and previous-purchase values.
- Confirmed no categorical whitespace issues.
- Removed promo_code_used after confirming it duplicated discount_applied.
- Confirmed 0 fully duplicated rows.
- Produced a cleaned dataset with 3,900 rows, 19 columns, and 0 missing values.

11. Cleaning Decisions and Assumptions

- The raw dataset is preserved unchanged; cleaning is performed on a loaded working copy.
- Category-level median imputation is used only for Review Rating because its missingness is limited and the affected rows retain useful information.
- purchase_frequency_days is an approximate analytical representation of categorical purchase frequency, not a measured interval between transactions.
- Quantile-based age groups are derived for segmentation while the original numerical age field is retained.
- No rows are removed because no confirmed full duplicates or documented invalid numerical records were found.

12. Output

The cleaned dataset is saved separately as customer_shopping_behavior_cleaned.csv. Final validation confirms 3,900 rows, 19 columns, 0 missing values, and 0 fully duplicated rows. The cleaned data is ready for downstream analysis while the original raw file remains unchanged.